Bowen Yang

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Research Interests

Algebra and mathematical physics, particularly commutative algebra in exactly solvable quantum lattice models. Homotopy-theoretic structure of quantum cellular automata and topological phases of matter. Mathematics of quantum error-correcting codes.

Employment

Postdoctoral Fellow, Harvard University	2024—present
Lecturer, Department of Mathematics, Harvard University	Spring 2027
Teaching Fellow, Department of Mathematics, Harvard University	Spring 2026

Grants and Awards

AMS-Simons Travel Grant, American Mathematical Society	2026—2028
Scott Russell Johnson Graduate Dissertation Prize, Caltech	2023
Rufus B. Kellogg Graduate Fellowship ($30,000 per annum)	2018—2021
Amherst International Student Scholarship, Amherst College	2014—2018

Publications

Published and Accepted

1. Operator algebra and algorithmic construction of boundaries and defects in (2+1)D topological Pauli stabilizer codes (with Zijian Liang, Joseph T. Iosue, Yu-An Chen), arXiv:2410.11942. *Accepted* **PRX Quantum** (2026).

2. A mathematical theory of topological invariants of quantum spin systems (with Adam Artymowicz, Anton Kapustin), **Comm. Amer. Math. Soc.** (2026), arXiv:2410.19287.

3. Transition between 2D Symmetry Protected Topological Phases on a Klein Bottle (with Vibhu Ravindran, Xie Chen), **Phys. Rev. B** (2026) arXiv:2510.00587.

4. Clifford quantum cellular automata from topological quantum field theories and invertible subalgebras (with Meng Sun, Zongyuan Wang, Nathanan Tantivasadakarn, Yu-An Chen), **PRX Quantum** (2026) arXiv:2509.07099.

5. Categorifying Clifford QCA, **Comm. Math. Phys.** (2026), arXiv:2504.14811.
6. Homological invariants of Pauli stabilizer codes (with Blazej Ruba), **Comm. Math. Phys.** (2024). arXiv:2204.06023
7. Spatial decay of Kubo's canonical correlation functions, **Lett. Math. Phys.** (2021). arXiv:1912.10831
8. A classification of invertible phases of bosonic quantum lattice systems in one dimension (with Anton Kapustin, Nikita Sopenko), **J. Math. Phys.** (2021). arXiv:2012.15491
9. Toy Teichmüller spaces of real dimension 2: the pentagon and the punctured triangle (with Yudong Chen et al.), **Geom. Dedicata** (2018). arXiv:1704.04331
10. "Strange" combinatorial quantum modular forms (with Amanda Folsom et al.), **J. Number Theory** (2017).

Preprints

1. Algebraic Theory of Abelian Fractons and Beyond (with Wilbur Shirley, Evan Wickenden, Agnès Beaudry, and Michael Hermele), arXiv:2610.40596.
2. Three-Dimensional QCA That Are Not Stably Circuits, arXiv:2610.40235.
3. Invertible Subalgebras for Chiral Semion Topological Order (with Yu-An Chen, Meng Sun, Nathanan Tantivasadakarn, and Zongyuan Wang), arXiv:2610.18953.
4. Poincaré Duality and Quadratic Refinements over Laurent Rings (with Błażej Ruba), arXiv:2609.25767.
5. Non-Clifford Quantum Cellular Automata from Invertible Topological Quantum Field Theories (with Meng Sun, Zongyuan Wang, Nathanan Tantivasadakarn, and Yu-An Chen), arXiv:2607.21697.
6. K-Theoretic Obstructions to Linearizing QCA Representations (with Mattie Ji), arXiv:2606.19657.
7. Resolving Spurious Topological Entanglement Entropy in Stabilizer Codes (with P. Han, Z. Liang, Y. Wang, Y. Gu, Y. Chen), arXiv:2604.27053.
8. The Classification of Pauli Stabilizer Codes: A Lattice and Continuum Treatise (with Matthew Yu), arXiv:2604.24847.
9. Quantum Cellular Automata: The Group, the Space, and the Spectrum (with Mattie Ji), arXiv:2602.16572.
10. Witt Groups and Bulk-Boundary Correspondence for Stabilizer States (with Blazej Ruba), arXiv:2509.10418.

Invited Talks

- 2027 *(forthcoming)*, **Caltech**, *Leinweber Symposium on Quantum Order*
- 2027 *(forthcoming)*, **University of Colorado Boulder**, *Center for Theory of Quantum Matter Colloquium*

- 2026 *(forthcoming)*, **New York**, *Simons Collaboration on Global Categorical Symmetries*
- 2026 *(forthcoming)*, **University of Notre Dame**, *Topology Seminar*
- 2026 *(forthcoming)*, **Johns Hopkins University**, *Topology Seminar*
- 2026, **Northeastern University**, *Applied and Interdisciplinary Mathematics (AIMS) seminar*
- 2026, **MIT**, *Ultra Quantum Matters Seminar*
- 2026, **Kavli IPMU, The University of Tokyo**, *Three lectures on Quantum Cellular Automata and Algebraic K-theory*
- 2026, **Stanford University**, *Quantum Cellular Automata Workshop*
- 2026, **MIT**, *Infinite-Dimensional Algebra Seminar*
- 2026, **La Jolla Meeting**, *Simons Collaboration on Global Categorical Symmetries*
- 2025, **University of Colorado Boulder**, *Topology Seminar*
- 2025, **Harvard University**, *CMSA Freedman Seminar*
- 2025, **New York University**, *Simons Collaboration on Global Categorical Symmetries*
- 2025, **MIT**, *Algebraic Topology Seminar*
- 2025, **University of Pennsylvania**, *Mathematical Physics Seminar*
- 2025, **University of California, Davis**, *Lattice Seminar* (online)
- 2024, **Amherst College**, *Math Colloquium*
- 2024, **Institute for Advanced Study**, *CMP/QFT Group Meeting*

Contributed Talks

- 2026, 2025, 2024, **Harvard University**, *Geometry and Quantum Theory Seminar*
- 2026, 2025, 2024, **Harvard University**, *CMSA Member Seminar*
- 2026, **San Juan, Puerto Rico**, *Algebraic Structures in Topology*
- 2025, **SLMath**, *Workshop on Infinite-Dimensional Division Algebras*
- 2025, **Caltech**, *Anton Kapustin's Group Meeting*

Education

Ph.D. in Mathematics, California Institute of Technology 2018—2023
Advisor: Anton Kapustin

B.A. in Mathematics, *summa cum laude*, Amherst College 2014—2018

Visiting Master's Student, New College, University of Oxford 2016—2017

Teaching Experience

Senior Thesis Advisor, Harvard University Summer 2026–Spring 2027
Student: Luke Zhu

Instructor, Harvard University Spring 2027 *(forthcoming)*
Math 25 — Honors Linear Algebra and Real Analysis

Summer REU Mentor, Harvard University Summer 2026
Student: Vincent Costa

Instructor, Harvard University Spring 2026
Math 21B — Linear Algebra and Ordinary Differential Equations

Teaching Apprentice, Harvard University Fall 2024
Math 21A — Multivariable Calculus

Instructor, California Institute of Technology Fall 2021
Math 8 — Calculus for Problem Solving

Course Assistant, California Institute of Technology 2018—2022
Math 2/102 — Differential Equations Fall 2018
Math 3/103 — Introduction to Probability and Statistics Winter 2019
Math 5/105a — Introduction to Abstract Algebra Fall 2019
Math 5/105b — Introduction to Abstract Algebra Winter 2020
Math 5/105c — Introduction to Abstract Algebra Spring 2019
Math 110c — Harmonic Analysis Spring 2020
Math 121a — Combinatorial Analysis Winter 2021
Math 121b — Combinatorial Analysis Spring 2021
Math 1c — Calculus Spring 2022
Math 120a — Commutative Algebra Fall 2022

Teaching Assistant/ Math Fellow, Amherst College 2015–2018
CS 101 — Python, Spring 2015
CS 121 — Java, Fall 2015
Math 350 — Groups, Rings, and Fields Fall 2015
Math 450 — Measure Theory and Integration Spring 2016
Math 355 — Introduction to Analysis Fall 2017
Math 405 — Lie Algebras Fall 2017

Counselor, Ross Mathematics Program Summers 2017, 2019

Professional Service

Co-organizer for *CMSA weekly Q and A Seminar*, 2024—2027.

Co-organizer for *Michael Freedman Seminar* from 2025—2027.

Referee for *SciPost Physics*; *Communications in Mathematical Physics*; *Quantum Computing Theory in Practice (QCTiP)*; *Quantum Information Processing Conference (QIP)*; *Qualitative Theory of Dynamical Systems*.